EE430 Laboratory Report 3: Dynamic Modelling and Simulation

Part 1: Simulating the Dynamic model of a 2 DoF Planar robot

Output:-

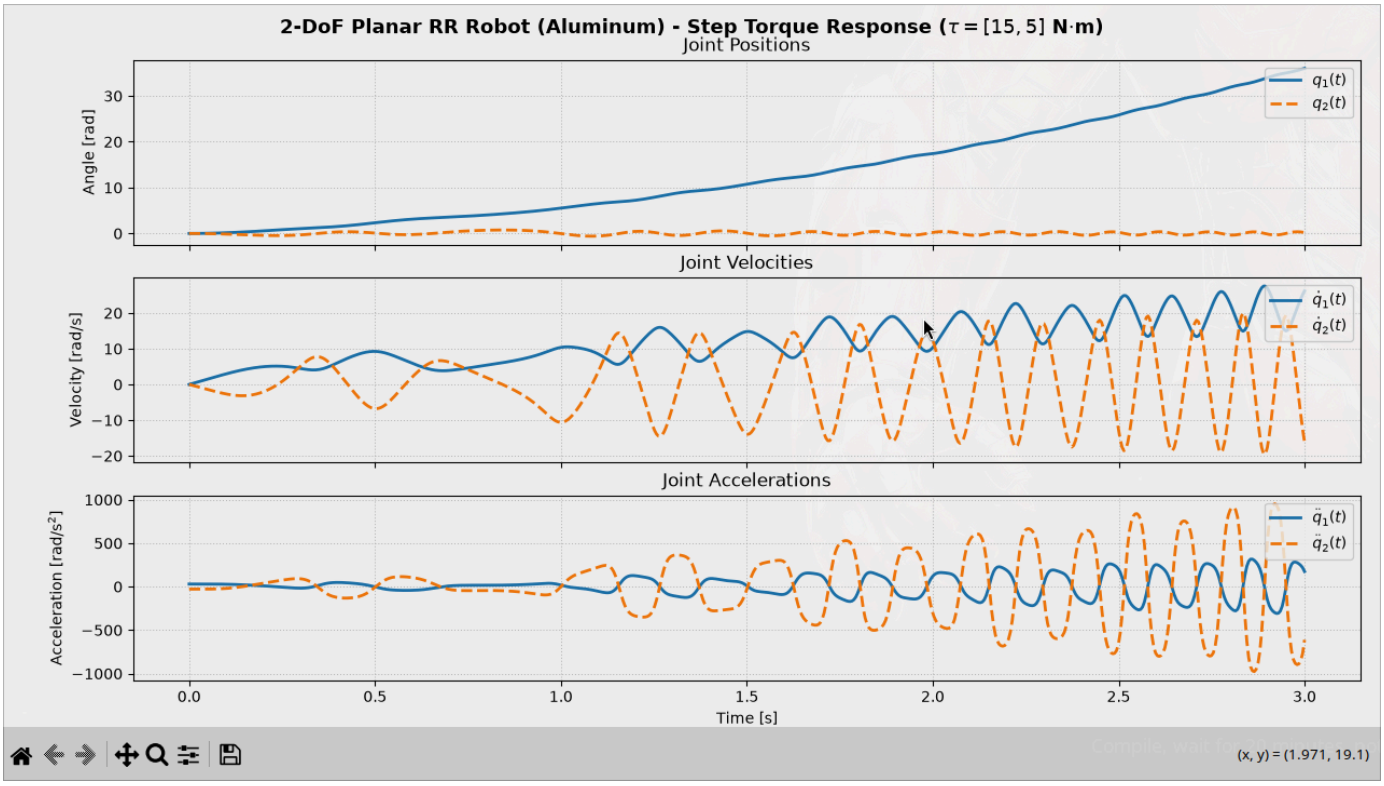


Fig 1. Step Response at both joints (q_1,q_2)

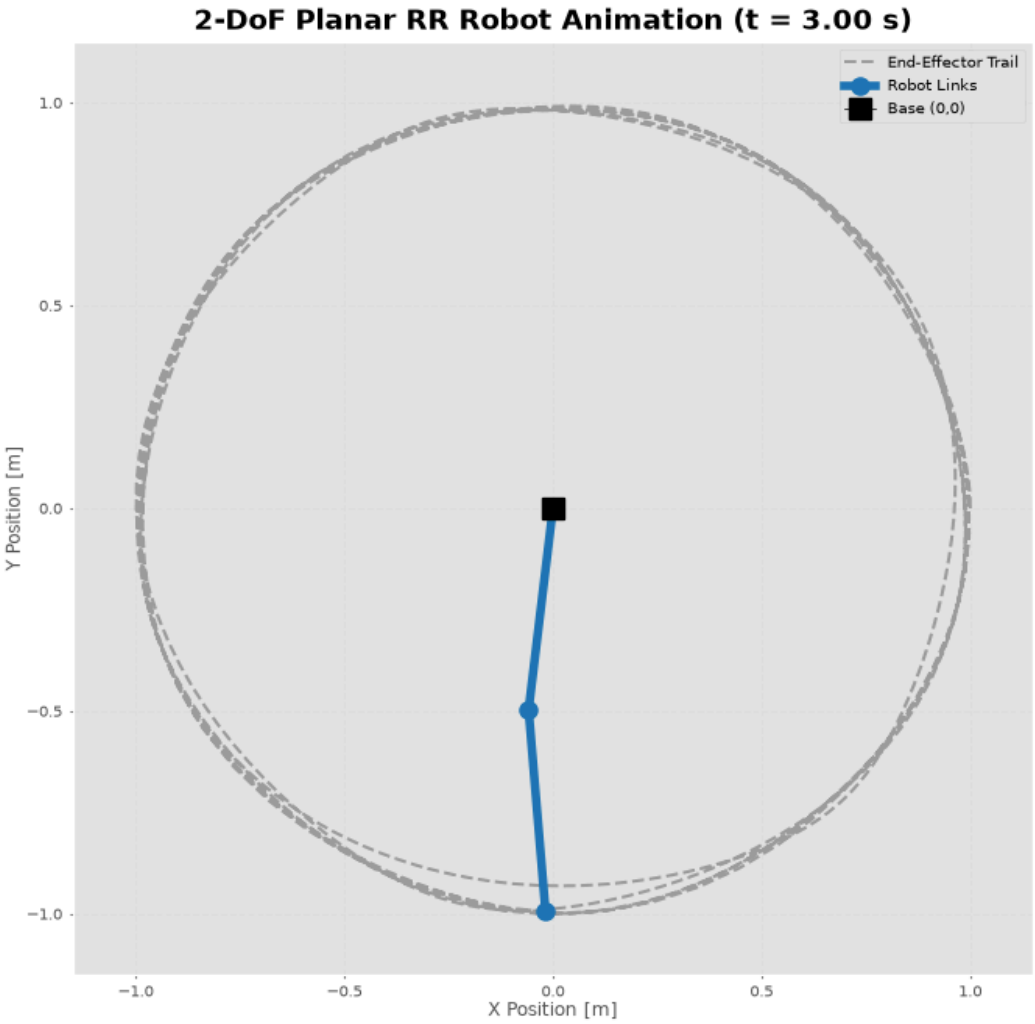


Fig 2. Stick Figure showing the movement of the robot (after t=3 sec)

Observation:-

After giving Step Torque, the end affector spins in circles. This causes the outer joint q_2 to wobble slightly, making its angle oscillate about zero, but the inner joint q_1 keeps rotating in the same direction, hence its position keeps increasing. Since there is constant energy being fed into the system, the velocities, and the accelerations keep increasing over time.

Part 2: Same task on a Yaskawa Motoman HC10

Output:-

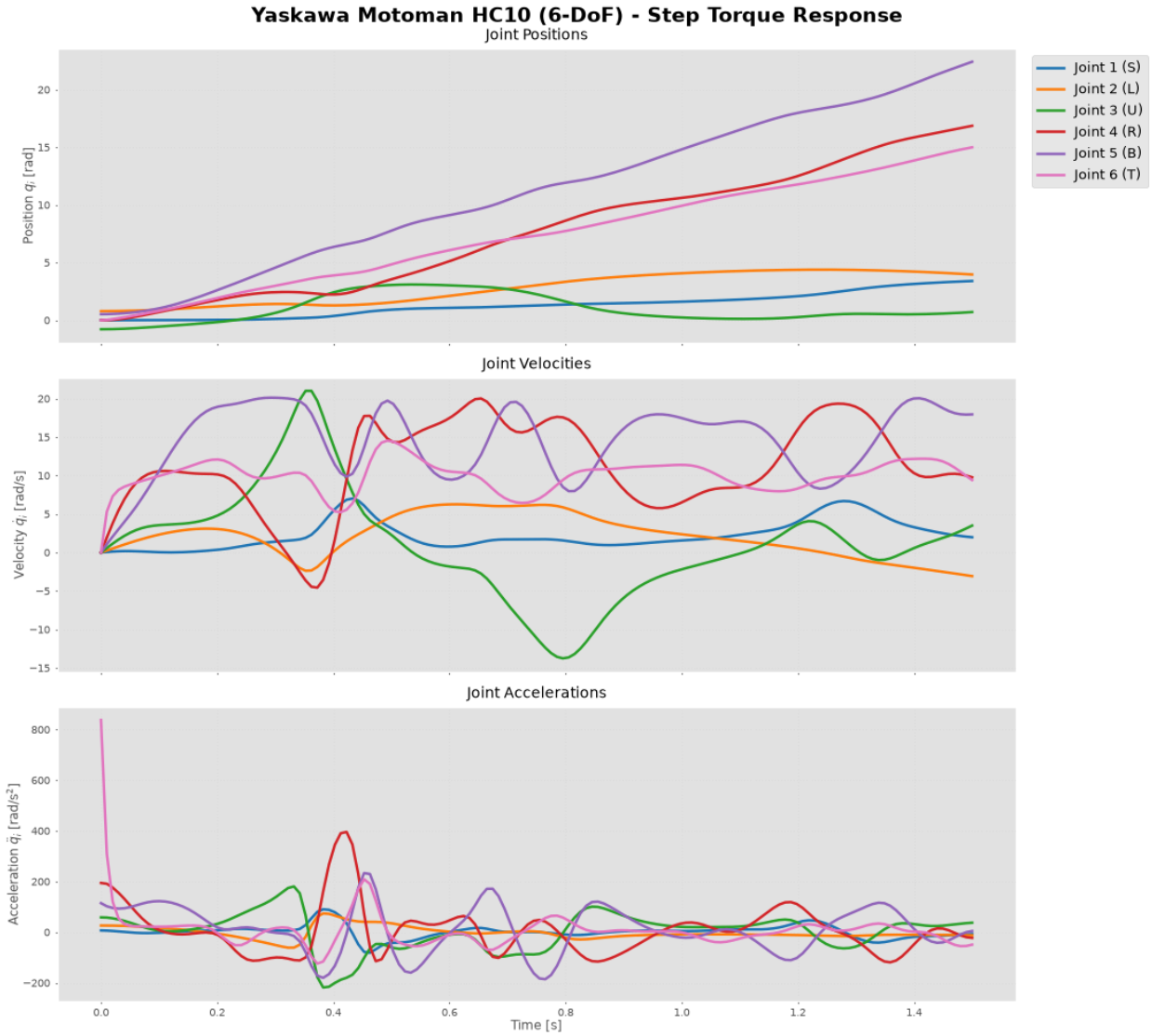


Fig 3. Step Response at all joints

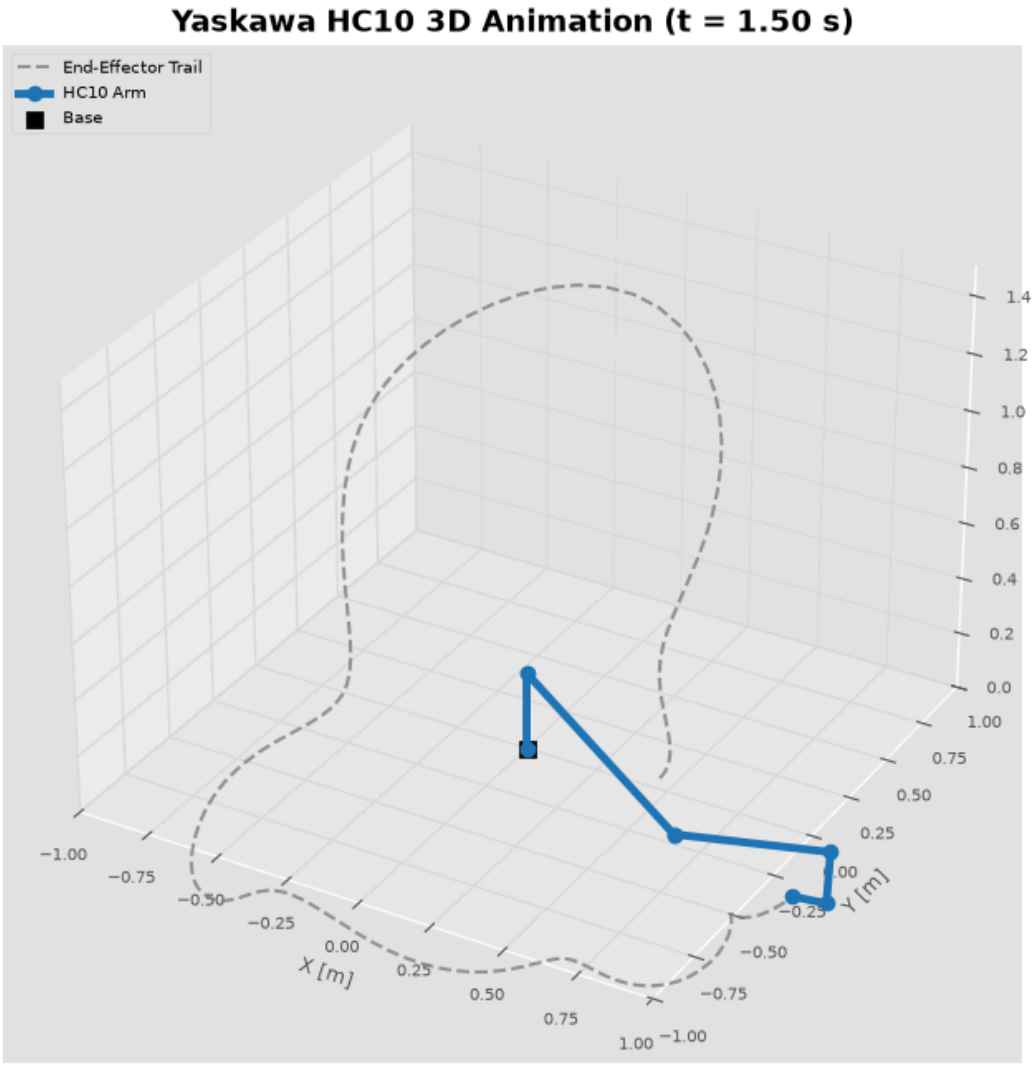


Fig 4. Stick Figure showing the movement of the robot (after t=1.5 sec)

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DHRobot: Yaskawa_Motoman_HC10, 6 joints (RRRRRR), dynamics, standard DH parameters
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θ_j	d_j	a_j	α_j
q1	0.275	0	90.0°
q2	0	0.7	0.0°
q3	0	0	90.0°
q4	0.5	0	-90.0°
q5	0.162	0	90.0°
q6	0.13	0	0.0°

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6x6 Inertia Matrix M(q) at q_home [kg.m^2]:
[[ 6.6756 -0.3499  0.4826 -0.1948 -0.0034 -0.0026]
 [-0.3499  9.3911 -0.3359 -0.0015  0.0342  0.    ]
 [ 0.4826 -0.3359  2.2386 -0.4826  0.0298  0.    ]
 [-0.1948 -0.0015 -0.4826  0.2008  0.0034  0.0026]
 [-0.0034  0.0342  0.0298  0.0034  0.0402  0.    ]
 [-0.0026  0.      0.      0.0026  0.      0.003 ]]
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Observation:-

After applying the step torques, the end-effector sweeps through a complex 3D swirling trajectory instead of a simple 2D circle. The base joint (Joint 1) and the lighter outer wrist joints (Joints 4, 5, and 6) continuously rotate in the direction of their applied torques, causing their positions to steadily increase over time. In contrast, the heavier arm joints (Joints 2 and 3) swing outward and oscillate/fluctuate due to strong gravitational resistance and dynamic cross-coupling from the moving links.